

Capstone Mela 2026

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Panel 1**Computer Vision, AI for Healthcare ,Modelling using
ML, LLMs, Industrial Data Analytics and NLP**

Team ID	Project Title
43	XAI powdered deep learning techniques for esophageal cancer detection using endoscopic images
47	Automated Radio Morphometric Analysis for Early Detection of Osteoporosis Using AI
51	Developing an advanced AI system that utilizes facial features and symptom patterns to predict rare diseases, aiming to revolutionize early medical diagnosis
79	Neuro control System Translating Brain Signals to motor Commands
162	MEDWISE – AI POWERED SMART HEALTHCARE ASSISTANT
38	AgentX: Automating AI Agent Creation through Knowledge-Driven Context Management
140	Price Forecasting and Profit Optimization for Retail and E-Commerce
141	Loop - A Next Gen Social AI Platform
145	Action Unit-Driven Emotion Recognition for Smart Home IoT Systems
160	Cost Estimation Tool for Manufacturing Units
94	A Blockchain-Secured Framework for Real-Time Threat Recognition in public safety domain Leveraging Multimodality using Deep Learning Fusion Models
98	Integrating Blockchain and Deep Reinforcement Learning for Secure Drug Traceability and Inventory Management in the Pharmaceutical Supply Chain

Panel 2

Applied ML, Gen AI and LLMs AND Cognitive Computing, Next Gen AI and ML Application

Team ID	Project Title
6	Foetal and Maternal Monitoring Using Edge AI and Conversational AI
124	Detecting side effects of drug-drug interactions
8	Forecasting Natural Disasters through Animal Behaviour Patterns using Deep Learning
24	Leveraging multi agent systems and vision language models for informed urban planning with community participation
36	Aerial Waste Surveillance using YOLOv8 with Multilingual SLM-Based Decision Support Framework
55	Building a Distributed Orderbook
75	Capstone Tracker With an Integrated Evaluation System
77	Exploring Optimization and Data Quantization for Edge Computing-tinyML
82	HyperCluster: Decentralized LLM Interface over Internet-Scale Peer-to-Peer Networks
85	Data Driven Football Formation and Transfer Recommendation System
117	Virtual Information Desk
139	Digital Twin for Computational Finance : An Adaptive Financial Model for Real-Time Forecasting and Decision Support
146	Multimodal YOLOv8 Perception and Bi-Level Adaptive Routing for Fairness-Aware Disaster Response

Panel 3**Multimodal AI in Action: Solving Real-World Problems
with Next-GenAI**

Team ID	Project Title
27	Multimodal Speech Therapy and Conversation Assistant in Indian Languages for Speech Impaired Individuals
31	Assistive Aid for Dyslexic Individuals using Real-Time Text Detection, Simplification and AR Integration
108	Decoding human brain activity to generate multimedia content to visualize thoughts for patients with LiS
128	AI Tutor for Dyslexic Children
41	AI-driven Agricultural Advisory system for Enhancing Rural Farmer Support with Indic Languages
3	Ecosystem simulation with real-time poaching sensing for Amazon Rainforest.
95	Multi-Language Bidirectional Sign Language Transfer for Video Conferencing using Machine Learning
116	Smart Footpath Connectivity System: Leveraging Image Segmentation and Crowdsourced Data for Urban Safety
143	A Holistic Approach to Manipulative Propaganda Detection on Social Media using Emotion Analysis with and GAN with a Sub-continental focus
148	Automated Vehicle Detection and Monitoring (AVDM) Intelligent Proximity Estimation for Autonomous Vehicles
154	Dynamic 3D Procedural Content Generation through Meta-Learning for Adaptable and Generalized Scene Creation
161	Integrated AI system for Behavior Analysis and Object Detection in Public Surveillance
92	Detecting Poisoning Attacks in Machine Learning Pipelines through Feedback-Driven Real-Time

Panel 4**Big Data, ML Applications, Blockchain and Gen AI**

Team ID	Project Title
50	XAI-MedTrack : Explainable AI for Longitudinal Medical Image Analysis on MRI Scans for Alzheimer's Diseases
29	IoTShield: A Novel Framework for Unsupervised Anomaly Detection and LLM Powered Triage in Industrial Control Systems
65	Analysis of Multi-Pollutant Particles Geospatial Visualization and Health Risk Assessment
93	Implementing Inertial Navigation System in Wearable Devices using Machine Learning
113	Automating Mobile Forensics Reporting
118	Automated Toolkit for Legacy Modernization Solutions
167	A Hybrid Federated and Centralized Learning approach for detecting fake Images.
49	Detection and Mitigation of Replay Attacks in CCTV System
52	Detecting Malicious User Activity by Correlating Digital Communications and System-Level User Behavior
63	Leveraging Machine Learning for Cryptanalysis: Identifying Cryptographic Algorithms Behind Ciphertexts
107	Content – Based Phishing Detection using NLP and DL
111	Privacy-Preserving Adaptive Authentication for Decentralized Telemedicine Systems
135	Federated Learning for fraud detection across financial institutions
138	Vendor Profiling and Behaviour Analysis in Dark Web Market Places

Panel 5

Applied ML and GenAI

Team ID	Project Title
14	Breast Cancer Detection Using Thermal Imaging
17	Smart AI Powered Post-Care Diabetes Management System
70	Enhance Autonomous Elderly Care
155	Real-Time Human Distress Sound Detection Using Edge-Based YAMNet Transfer Learning
15	Characterization and Detection of Diseases in Sugarcane Leaves
10	Cognify Learning Suite
42	NIGP-GS: A Bayesian Nonparametric Framework for Hybrid-Query Processing on Graph Streams
67	Metahuristic and Machine Learning Based Approach for Load Balancing and Resource Optimization
97	Real-Time Breathing and Posture analysis for Pranayama
120	Weapon and Pose Alerting System
123	AI-Driven Tribunal Case Management
88	A Hybrid Approach to detect Illegal Activities in Dark Web Data
132	Graph Based Botnet Detection and Automated Network Defense Using GNN and SDN

Panel 6**Applied ML in Medical, Software Testing & Networks
and LLM attacks**

Team ID	Project Title
2	Early Prediction and Classification of Cervical Cancer Using Deep Learning
9	Pathway-Driven Multi-Omics Graph Framework for NAFLD Prediction
105	Colorectal Polyp Detection Using SAM-HQ: A zero-Shot Learning Approach for Medical Image Segmentation
106	Predictive Maintenance of Medical Equipment Failure Through Real-Time Data
114	Detection of Multiple Neurological disorders using Machine Learning and Deep Learning
126	AI-Powered Skin Care Assistant for personalized and Ethical Product Selection
129	Detection of Paediatric Pneumonia Using Chest X-Ray's
32	Mindful Motion: Siddha-Based ML Body Classification , Mudra Posture Correction, and Chatbot for Holistic Wellness
69	Developing Adaptive Self Driving cars using LNN and RL for Indian roads
78	"Echoes of Vanishing Voices: Reviving Endangered Languages"- focusing on Konkani and Lambani
122	Automatic Unit Testcase Generation Using Large Language Models (LLM's)
157	Advancing Lip-to-Speech synthesis
19	Multi-Agent Framework for the prevention of LLM Jailbreak Attacks
152	SmartGuard: A Machine Learning Based Classification Framework for Automated DoS/DDoS Defence in Next Generation Firewall Systems

Panel 7**Digital Vision and multidisciplinary (IOT, Security, ML)**

Team ID	Project Title
34	A Framework for agricultural supply chain management aligning crop production with market demand
4	Optimizing Recommender Systems: A Holistic Approach with Market Basket, Segmentation, and Sentiment Insights
35	Anticipating Hate: A Predictive Framework for Online Hate Speech
48	Smart Pill Dispenser with Vitals Monitoring and Prescription Management
58	Automated Incident Response System Using Zero Trust Architecture
74	AR for Glare Light Tracking and filtering using deep learning
87	Reviving Karnataka's Farmlands: AI-driven Biodiversity Conservation for Sustainable Growth
99	Predictive Analysis of Communication in Industrial IoT (IIoT) Systems Using Machine Learning
100	JustiTech- AI Enhanced Legal Judgement Framework
101	Investigation and Conservation of Handwritten images to speech and braille
134	An Intelligent Chord Progression Analyser for Indian Music
159	Augmented Reality for pothole Detection System for New-age Vehicles
169	CoreWatch: AI-Driven CPU/GPU Performance Analyzer
83	Integration of software Defined networking and Machine Learning techniques in detection and mitigation of DDoS attacks
127	Energy Efficient Hybrid Intrusion Detection System for New-age vehicles

Panel 8**SDN, Blockchain, IoT, Cloud Computing, Deep Learning
and Machine Learning Applications**

Team ID	Project Title
23	AI Powered Virtual Home Architect
26	RSSI Fingerprinting-Based Distributed Computing Framework
170	Intelligent Concept Acquisition System
11	Development of Hybrid Consensus Algorithm based Decentralization Smart Transportation System for V2V Communication
22	Kubernetes Scheduler for High Availability Topology to Reduce Latency in Microservice Orchestration
28	Decentralized WebAssembly-Powered Distributed Computing Framework
57	Design And Development of an FPGA based quantum cryptography protocol for secure communication
110	Combat Readiness and Crisis Response System, Powered by Federated Learning and Blockchain
121	Enhancing Security and Real Time Analysis in RPM devices using Blockchain and ML Models
142	BioMimetic Cyber Defence Matrix – Adaptive Deception and Swarm Intelligence for Advanced Threat Mitigation
151	Unsupervised Ensemble Learning for Real-Time Detection of Zero-Day Vulnerabilities in Dynamic Networks
171	Deep Learning-Driven Image Based-Malware Detection for IoMT Security

Panel 9

Big Data, Digital Twin, Blockchain and Smart Networks

Team ID	Project Title
44	Phonocardiogram-Based System for Enhanced Detection of Congenital Heart Diseases in Children
112	Real-Time Mental Wellness AI Companion
33	Leveraging MLLM's on FOG in Smart Agriculture
56	LLM-Driven AI Framework for Company Performance Prediction
64	CheatGuard: IoT and ML Powered Smart Exam Surveillance
66	StepUP.ai
149	GitHired: AI-Enhanced Campus Recruitment Management System with Natural Language Data Intelligence
5	Advanced IoT-Driven Real-Time Tracking Through LoRa Telemetry and Cloud Stream Analytics
16	LLM Based Cyber Threat Detection and Mitigation for Microservices
25	Leveraging Uni-kernels to Enhance Performance and Scalability of Blockchain Systems
89	Quantum Resilient Cybersecurity for Avionics Networks : A Cloud Native, API Driven Solution Integrating Blockchain
102	SCADA System for Real-Time Monitoring and Cyber-Threat Detection in Electrical Substations
147	Misbehaviour Detection using Deep Learning in VANET's
164	Real-Time Plagiarism Detection and Software Forensics Using Activity Logs and Big Data
168	Enhancing Password Guessing Efficiency- A Clustered Approach Beyond PGTCN

Panel 10**Leveraging machine intelligence in diverse fields**

Team ID	Project Title
60	AI-Driven Real-Time Analysis of Amniotic Fluid Index for Fetal Health Assessment and Decision Support
62	Deep Learning for Facial Skin Condition Detection with personalized Chemical Component Recommendation
81	Early Detection of Myocardial Infraction using Advanced AI Techniques
104	Osteoporosis Detector with Explainable AI
125	An Interpretable Model for Stroke Prediction and Arrhythmia Classification via ECG and PPG
131	Retinal Image Characterization using vascular feature extraction
91	AgriRAG: AI Driven Platform for empowering Farmers with Policy Insights and Financial Support
1	AI-Based Resume Screening
68	Fortifying digital medical images security through hybrid authentication and embedding techniques
73	Product Insights and Transparency Architecture for Data-driven Analysis
86	AN Integrated Framework for Predicting Microplastic Pollution and Assessing Human Health Risks
144	AVIOS – Adaptive Vigilant Intelligent OS
163	Multimodal Analytics for Real-time Sports Narration

Panel 11**AI vision and model intelligence**

Team ID	Project Title
30	Predictive analysis of water, quality and waterborne diseases risk using machine learning
72	Multi Modal Prediction of Coronary Artery Diseases
80	Automated BCS Drug Categorization Using Machine Learning: An Explainable AI Approach for BCS Classification System
21	Satellite Based Fair Compensation for Crop-Loss
46	Real-Time Deception Detection Using Micro-expression Analysis with Audio-Visual Cues
54	Real Time Fraud Call Detection Using Machine Learning
90	Conversational AI for Legal Consultation and Compliance with emphasis on Real Estate Law in India
96	AI-Enabled Predictive System for Aircraft Maneuvers Across Aircraft Types
115	SocialSense: A Neuro Symbolic Approach for Contextual Social Reasoning and Guidance
119	Detecting and Enhancing the Operational Efficiency of ADAS during "WRAITHGLOOM" condition
153	Behavioural and Biometric Insights Based Authentication of Dynamic Signatures
165	Bird Speech Classification using multimodal data and migration forecasting

Panel 12**Applied Intelligence, Healthcare, and AgriTech Analytics**

Team ID	Project Title
13	Silent Speech Recognition Using Electromyography and LLM's: A Multi-Modal Approach
59	Enhancing Successful aging: A holistic approved approach to well-being
137	Realtime detection of Deepfake Manipulations in Medical
156	Comparative Analysis of Quantum Entanglement Techniques for Parkinson's Disease Detection: Evaluating Encoding Strategies in Quantum Machine Learning
166	Multimodal Intent Prediction for Autistic Individuals
20	Plant Pal – Drone and NLP System for Intelligent Plant Diseases Diagnosis and Improved User Engagement
7	Enhancing Electric Grid Reliability Through Machine Learning and Deep Learning and Deep Learning Based Fault Detection
12	PESUPrep
18	UNMASKING GST FRAUD: Detecting HSN Code Misuse, Circulating Trading and B2C Sales Suppression
71	Fusion Flavors: AI-Driven Cross-Culture Recipe Generator For Personalized and Interactive Culinary Experiences
76	Personalized Group and Individual Recommendation System for Domain Specific Research Articles and Authors
133	CODEFEST: AI-Driven Code Completion and Prompt-Driven Code Generation for C, Go and Python Programming
150	Predicting Budget Allocations Using Sectoral Growth: An Empirical Assessment of Deep Learning vs Linear Models for India's Sectors

Panel 13**Intelligent Computing Systems for signal Processing
and diverse applications**

Team ID	Project Title
37	Automated Evaluation and Visualization of Rehabilitation Exercises using Body Pose Landmarks and Features Focusing on Tennis Elbow
39	Early detection of sleep apnea
45	Early Detection of Osteoporosis using Dental X-rays
53	Automated Screening and Grading of Diabetic Retinopathy Using Hybrid Deep Learning Models
136	Mental Health Misinformation Detection in Social Media Using LLMs and Knowledge Graphs
40	Audio to sign language Translation system using 3d avatar for ISL
61	Detecting Cognitive and Publication Bias in Scientific Literature Using Advanced NLP Techniques
84	Drivewise: Your attention assistant
103	Neural Narratives: Thought Driven Speech Synthesis
109	Conversational AI driven ML model for cyber threat identification and mitigation in Smart factories
130	Data-Driven Influencers Selection for Optimized Sponsorship in Marketing Campaigns
158	Social Media Bot Detection with Sentiment Analysis